MIL-C-60593 (MU)
20 September 1966
SUPERSEDING
FA-PD-MI-2670
10 June 1965

MILITARY SPECIFICATION
CUTTER, POWDER ACTUATED, M8
RE-MANUFACTURED METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers re-manufactured Cutter, Powder Actuated, M8, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATION

Military

MIL-A-2550	-	Ammunition and Special Weapons, General Specification For

STANDARDS

Federal

FED-STD-141	-	Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-413	-	Visual Inspection Guide for Rubber 0-rings
MIL-STD-639	-	Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	-	Ammunition Data Card

FSC 1377

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2.1 (cont)

DRAWINGS

Munitions Command

F10520567	-	Cutter, Powder Actuated, M8, Metal Parts Assembly
C8835919	-	Container, Ammunition, Fiber M436 for Cutter, Cartridge Actuated, M8 with Cartridge M122
C8835920	-	Box, Packing, Ammunition for Cutter, Cartridge Actuated, M8 with Cartridge M122 in Fiber Container M436
IEL-10520567	-	Index of Inspection Equipment List for Cutter, Powder Actuated, M8, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing Materials - D149-61 - Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Material.- The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intented for use in manufacture.

3.3 Physical properties.- Physical properties required by the drawing or specification shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

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3.4 Hydrostatic pressure.- The body and swivel shall withstand a constant internal pressure of 1500 pounds per square inch (psi), minimum, for 15 seconds, minimum, without application of any external load other than that applied by the liquid pressure and shall show no evidence of leakage or mechanical failure. (See 6.1)

3.5 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing.

3.5.1 The re-manufactured cutter metal parts assembly shall contain a new shear pin, shear unlock pin, set screws, 0-rings, cableholder clip, gasket, cotter pin and car seal.

3.5.2 Torque.- Torque requirements shall be as specified in table I.

Table I

Component	A Run On, Min (inch pounds)	B Seating or Bottoming (Including A) (inch pounds)
Cap and swivel	40	300 + 15
Head and body	40	*

* Not required. Head shall be run on body to a position where initial bottom contact is made and then no further torquing need be performed.

3.5.3 Swivel rotation.- Upon completion of assembly, the swivel shall be free to rotate 360 degrees.

3.6 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.7 Blade coating.- The requirements for blade coating are as specified on the applicable drawing and the following:

3.7.1 Electrical Insulation.- The coating shall be capable of withstanding 1000 volts AC without suffering puncture, charring or any other physical damage.

3.7.2 Solvent resistance.- The coating shall show no evidence of peeling, blistering, dissolving or loosening when subjected to acetone exposure.

3.7.3 Appearance.- The coating shall be free from blisters, cracks, wrinkles or other irregularities.

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3.8 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than three degrees in either direction when passed in front of the compass at a distance of six inches.

3.9 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.9.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.9.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- The shipping lot of metal parts assemblies shall consist of 318 cutter, maximum, provided the homogeneity requirement of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

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4.2.2 Examination.- The ordnance standard finish blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.2.2.1 Re-claimed parts.- Examination of re-usable parts, which require no re-processing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be re-processed or rejected.

4.2.2.2 Re-processed parts.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in tables II and III. For an individual defect (single characteristic) the criteria in table II shall be used. For total defects within a major or minor group, the criteria in table III shall be used. All nonconforming material shall be rejected.

Table II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1,300	110	1	2	2	3
1,301 - 3,200	150	2	3	3	4
3,201 - 8,000	225	3	4	4	5
8,001 - 22,000	300	4	5	5	6

Table III

Total Allowable Defects per Sample Size

Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14

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4.2.2.2 (cont)

Sample Sizes for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.2.2.3 Body Assembly, Dwg F10520569 covering a detail of Dwg F10520567 (prior to inserting pellets).

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
101. Pitch diameter of long thread, min	Gage
102. Major diameter of long thread, min	Gage
103. Pitch diameter of short thread, min	Gage
104. Major diameter of short thread, min	Gage
105. Thread damaged or not full	Visual
106. Metal defective	Visual
107. Protective finish inadequate or defective	Visual
108. Surface finish of large and small bores, improper	Visual
109. Burr or sharp corner	Visual
110. Foreign matter	Visual
MINOR	
201. Surface finish improper, except as otherwise classified	Visual

4.2.2.4 Body Assembly, Dwg F10520569 covering a detail of Dwg F10520567. (After inserting pellets).

CRITICAL	
None defined.	
MAJOR	
101. Pellet missing, loose or improperly inserted	Manual and Visual
102. Thread damaged	Visual

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4.2.2.4 (cont)

	Defect	Method of Inspection
103.	Burr	Visual
104.	Foreign matter	Visual

MINOR
None defined.

4.2.2.5 Blade, Dwg D10520572 covering a detail of Dwg F10520567.

CRITICAL
None defined

MAJOR

101.	Cutting edge, flat	Visual
102.	Metal defective	Visual
103.	Protective finish inadequate or defective	Visual
104.	Surface preparation, improper or inadequate	Visual
105.	Surface finish of blade, improper	Visual
106.	Burr or sharp edge, except as otherwise classified	Visual
107.	Foreign matter	Visual

MINOR

201.	Coating improper, inadequate or defective	Visual
202.	Surface finish improper, except as otherwise classified	Visual
203.	Marking missing, incorrect or illegible	Visual

4.2.2.6 Head, Dwg D10520573 covering a detail of Dwg F10520567.

CRITICAL
None defined.

MAJOR

101.	Pitch diameter of large internal thread, max	Gage
102.	Minor diameter of large internal thread, max	Gage
103.	Pitch diameter of gas port thread, max	Gage
104.	Minor diameter of gas port thread, max	Gage
105.	Thread damaged or not full	Visual
106.	Metal defective	Visual
107.	Protective finish inadequate or defective	Visual

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4.2.2.6 (cont)

Defect	Method of Inspection
108. Surface finish of long cavity and angle of gas port, improper	Visual
109. Burr or sharp corner	Visual
110. Foreign matter	Visual
MINOR	
201. Surface finish improper, except as otherwise classified	Visual

4.2.2.7 Pin, Firing, Dwg C10520574 covering a detail of Dwg F10520567.

CRITICAL
None defined.

MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr or sharp corner	Visual
104. Surface finish, improper	Visual
105. Foreign matter	Visual

MINOR
None defined.

4.2.2.8 Anvil, Dwg B10520575 covering a detail of Dwg F10520567.

CRITICAL
None defined.

MAJOR	
101. Metal defective	Visual
MINOR	
201. Protective finish inadequate or defective	Visual
202. Burr or sharp edge	Visual

4.2.2.9 Cap, Dwg C10520578 covering a detail of Dwg F10520567.

CRITICAL
None defined.

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4.2.2.9 (cont)

Defect	Method of Inspection
MAJOR	
101. Pitch diameter of thread, max	Gage
102. Minor diameter of thread, max	Gage
103. Thread damaged or not full	Visual
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Burr or sharp edges	Visual
107. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.10 Nut, Mounting, Dwg B10520579 covering a detail of Dwg F10520567.

CRITICAL	
None defined.	
MAJOR	
101. Pitch diameter of thread, max	Gage
102. Minor diameter of thread, max	Gage
103. Thread damaged or not full	Visual
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Burr or sharp edges	Visual
107. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.11 Swivel, Assembly, Dwg F10520581 covering a detail of Dwg F10520567. (Prior to inserting pellets).

CRITICAL	
None defined.	
MAJOR	
101. Pitch diameter of thread, min	Gage
102. Major diameter of thread, min	Gage
103. Thread damaged or not full	Visual

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4.2.2.11 (cont)

Defect	Method of Inspection
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Surface finish of body, large bore and angles, improper	Visual
107. Burr or sharp corners	Visual
108. Foreign matter	Visual

MINOR
None defined.

4.2.2.12 Swivel Assembly, Dwg F10520581 covering a detail of Dwg F10520567. (After inserting pellets).

CRITICAL
None defined.

MAJOR	
101. Protrusion of pellets, max	Gage
102. Pellet missing, loose or improperly inserted	Manual and Visual
103. Thread damaged	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Location of pellets	Gage

4.2.2.13 Cutter, Powder Actuated, M8, Metal Parts Assembly Dwg F10520567.

CRITICAL
None defined.

MAJOR	
101. Swivel not free to rotate 360 degrees	Manual
102. Blade shear pin, missing	Visual
103. Any part missing, except as otherwise classified	Visual
104. Improper assembly	Visual
105. Index hole in cap, missing	Visual
106. Lubricant missing or excessive	Visual
107. Marking for matching parts identification missing, incorrect or illegible	Visual

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4.2.2.13 (cont)

Defect	Method of Inspection
108. Burr	Visual
109. Foreign matter	Visual
MINOR	
201. Barrier material in set screw, missing	Visual
202. Shipping plug missing	Visual
203. Locking wire missing	Visual
204. Protective finish inadequate or defective	Visual

4.2.2.14 Container, Fiber, Dwg C8835919.

CRITICAL
None defined.

MAJOR
None defined.

MINOR	
201. Gap between body and cover, max	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual

4.2.2.15 Box, Packing, Dwg C8835920.

CRITICAL
None defined.

MAJOR
None defined.

MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual
204. Contents exposed	Visual

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4.2.2.15 (cont)

Defect	Method of Inspection
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.3:

4.2.3.1 Hardness.- Fifty units from each heat-treated lot of components shall be tested for compliance with the values noted on the applicable drawing. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot of components.

4.2.3.2 Hydrostatic pressure.- Each body and each swivel shall be subjected to an internal hydrostatic pressure. Failure of any unit to comply with the requirement of 3.5 shall be cause for rejection of the unit.

4.2.3.3 Shear pin strength.- Fifty pins of each type from each lot shall be tested for shear strength. Failure of any shear pin to comply with the requirements of the drawing, as applicable, shall be cause for rejection of the lot.

4.2.3.4 Torque.- Each cap and swivel subassembly shall be subjected to the torque application specified in table I, 3.5.2. Failure of any subassembly to produce the requirements specified shall be cause for rejection of the subassembly.

4.2.3.5 Protective finish.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable military standard listed on the drawing.

4.2.3.6 Blade coating.

4.2.3.6.1 Eight blades from each lot of blades shall be tested for thickness, electrical insulation and solvent resistance. Failure of any unit to comply with the requirements of the drawing and 3.7.1 through 3.7.3, as applicable, shall be cause for rejection of the lot.

4.2.3.7 Residual magnetism.- Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit to comply with the requirement of 3.8 shall be cause for rejection of the assembly.

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4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10520567.

4.3 Test methods and procedures.

4.3.1 Hardness.- Prior to the application of protective finish, a hardness testing machine shall be used to determine the hardness values specified on the detail drawing.

4.3.2 Hydrostatic pressure.- The test shall be performed prior to application of protective finish. The body and swivel, individually, shall be mounted in a device and subjected to the hydrostatic test in accordance with the requirement of 3.4.

4.3.3 Shear pin strength.- After application of protective finish, a load shall be applied at the steady rate of 0.125 inch per minute.

4.3.4 Torque.- The torque shall be applied to the cap, prior to drilling of hole in the cap and assembly of the lock wire.

4.3.5 Protective finish.- Test procedures for protective finish shall be in accordance with the applicable military standard referenced on the drawing.

4.3.6 Blade coating.

4.3.6.1 Thickness.- Test procedures shall be performed in accordance with Federal Test Standard Number 141, Method 6181, except that the blades shall be used in lieu of test specimens.

4.3.6.2 Electrical insulation.- The electrical insulation test shall be performed in accordance with ASTM Standard - D149-61, except that an electrode of 1/4-inch diameter rounded to a 1/8-inch radius shall be used, AC voltage to be impressed at the rate of 250 to 300 volts per second, until 1000 volts are reached at which time the electrode is immediately removed from the coating. The coating is then visually examined to determine compliance with the requirement of 3.7.1.

4.3.6.3 Solvent resistance.- The coated part shall be rubbed lightly by hand for 30 seconds with a clean white cloth that has been saturated with acetone. Upon completing the test, the coating shall be air-dried and visually examined to determine compliance with the requirement of 3.7.2.

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4.3.7 Residual magnetism.- The cutter shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the cutter six inches from and at the same height as the compass, in a north-south horizontal position with the compass set in an east-west heading. The cutter shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90-degree rotation of the cutter.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Dwgs C8835919 and C8835920.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 When re-manufacturing and loading of an assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

6.3 Re-manufactured components are as follows: Head, Body Assembly (minus pellets), Mounting Nut, Blade, Anvil, Swivel Assembly (minus pellets), and Cap.

6.4 Nylon pellets shall be removed from the body assembly and swivel assembly and new pellets inserted.

6.5 The coating shall be stripped from the blade and re-coated after plating.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A100

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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